CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VIth Semester of B.Sc. Biotechnology/Microbiology/Biochemistry Examination May-2019

BE607 - BASIC FORENSIC BIOTECHNOLOGY

Date: 06/05/2019 Day: Monday Time: 01:30 pm To 01:50 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

1. ________________ is used for the development of fingerprint.
   - a. Sodium chloride
   - b. Sodium nitrate
   - c. Ninhydrin
   - d. Sodium hydroxide
2. __________________ evolved a mathematical formula to supplement Henry's idea of sorting slips based on fingerprint patterns.
   - a. Khan Bahadur Azizul Huq
   - b. Rai Bahadur
   - c. Hemchandra Bose
   - d. Rai Bahadur Azizul Huq
3. The first state forensic laboratory in India was established at ___________.
   - a. Shimla
   - b. Chandigarh
   - c. Hyderabad
   - d. Calcutta
4. Full form of NDPS is ______________________________________.
   - a. Narcotic Drugs and Psychotropic Substances
   - b. Normal Doses of Physiological Substances
   - c. Neutral Drugs and Poisonous Substances
   - d. National Database of Physiological Substances

5. _______________ instrument is used for the measurement of uncontrollable physiological reactions in Lie-detection division.

a. Cardiograph b. Physiograph
c. Polygraph d. None of the above

6. In forensic science, epiphyseal fusion helps to determine age group between ____.

a. 0-5 years b. 06-25 years
c. 25-40 years d. 40-60 years

7. D17Z1 probe is a very specific and consists of _____ to bind with chromosome 17.

a. 10 basepairs b. 20 basepairs
c. 30 basepairs d. 40 basepairs

8. Phenolphthalin assay is also known as a ______________ assay.

a. Kastle-Meyer b. Kriger-Meyer
c. Kaslin-Munaximib d. Kraiger-Munab

9. In presumptive assays, for recognition of blood, heme molecule acts as a/an _____.

a. oxidant b. reductant
c. biocatalyst d. substrate

10. ________________ system included morphological body descriptions.

a. Bertillon’s b. Goddard’s
c. Henry’s d. Galton’s

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Date: 06/05/2019 Day: Monday Time: 01:50 pm To 03:30 pm Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in TWO SEPARATE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show the necessary calculations.

SECTION -I

Q - II	**Answer the following questions as directed.**	**10**
Q-IIA	**Answer the following questions as directed [Any Two].**	**6**
1	What are CFSLs and SFSLs? Discuss in detail.	
2	Write a short note on role of document, chemistry and biological divisions of FSL.	
3	Describe Immunochromatographic and Enzyme-linked immunosorbent assays for identification of salivary amylase in detail.	
Q-IIB	**Write a short note on <u>any one</u> of the following:**	**4**
1	Define VNTR and STR. Provide differences between them.	
2	Explain microcrystal assays.	

SECTION - II

Q-III	**Answer the following questions as directed.**	**15**
Q-IIIA	**Answer <u>any two</u> of the following:**	**10**
1	Write a short note on “Laws of Forensic Sciences” and provide suitable examples.	
2	Mention vital informations for collection of bite mark evidences, types, analysis and comparison of bite marks.	
3	Explain DNA fingerprinting in detail and give its applications. Draw a labelled diagram.	
Q-IIIB	**Answer <u>any one</u> of the following:**	**5**
1	Draw different fingerprint patterns and explain in detail.	
2	Explain “SLOT-BLOT assay”. Mention colorimetric and chemiluminescent detection.	